

CARGOMULE · REQUIREMENTS

CargoMule requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Status from CGM-CAL-001 v0.6 with the 2026-10-02 decisions carried in: R8 met on paper at 46.3 kg; R12 now over the value-engineering target by \$4 (\$1,004); no target changed
0.7	2026-10-02	Amish Chadha	R8 restated as a 47 kg hard cap for the first prototype (CGM-DDR-003 A1, accepted 2026-10-02), now met on paper; private-ground rule for O2 noted
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target; cost wording only, no number changed
0.5	2026-10-01	Amish Chadha	Status from CGM-CAL-001 v0.3 for the constructable design (CGM-DDR-003); R8 now not met at 46.2 kg; no target changed
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002). R8 relaxed to 45 kg; R3 restated with thermal derating and a use limit; status from CGM-CAL-001 v0.2

0.3	2026-09-25	Amish Chadha	TRL 3. R8 relaxed to 40 kg and R12 redefined as the \$1,000 budget (CGM-DDR-001); status from CGM-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-24	Amish Chadha	Initial scaffold

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These requirements are checked by calculation in CGM-CAL-001 v0.6 at TRL 3, for the constructable design of CGM-DDR-003. Targets changed with Amish's decisions of 2026-09-25. Under CGM-DDR-001, R8 was relaxed from 35 kg to 40 kg for the steel frame (D3) and R12 was redefined against the \$1,000 prototype value-engineering target (D1). Under CGM-DDR-002, R8 is relaxed again to 45 kg for the first prototype, and R3 is restated: the 300 m climb target stands, and beyond it the controller derates motor current to protect the winding, with the use limit stated to riders. On 2026-10-02 Amish accepted the response to the R8 miss (CGM-DDR-003, A1): R8 is 47 kg as a hard cap for the first prototype only, and the trailer is weighed at TRL 4. Against the calculations, **three are at risk (R3, R6, R7)**; eight are met on paper or by design, R12 is over its value-engineering target by \$4 (a target, not a limit), and two cannot be verified at TRL 3. Targets are not yet validated with users; the first users and region for co-design are still open (CGM-DDR-001, O1).

The **design load case** is 150 kg of payload centered on the deck, the trailer (about 46 kg, CGM-CAL-001) and a towing bicycle with rider of about 100 kg, on a dry paved road. The **design route** is a 10 km round trip with 120 m of climbing, 120 m of descent and 20 stops.

Table 1. Requirements, targets and status at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CGM-CAL-001)
R1	Carry the payload	150 kg (330 lb) on a deck of at least 0.8 m ² , with tie-down points	Frame stress calculation; later proof load	Met on paper: deck 0.84 m ² ; least structural factor 1.6 at 2 g and the ultimate hitch case
R2	Range per charge	20 km or more on the design route in the design load case	Energy calculation; later logged ride	Met on paper: 24.5 km
R3	Assist on hills	8 % grade for 300 m at 8 km/h or more with the rider feeling 40 N or less of drawbar pull, without motor over-temperature at 30 °C. On longer continuous climbs the controller derates motor current so the winding stays at or below 100 °C; the use limit (about 300 m of continuous 8 % climbing at full assist) is stated in the rider instructions (restated, CGM-DDR-002)	Force and thermal calculation; motor datasheet at part selection	At risk: 38.7 N felt; winding about 96 °C against a 100 °C limit on assumed motor constants; derating starts after about 331 m [C10]
R4	Near-zero added load on the flat	Steady drawbar pull felt by the rider 15 N or less at 5 to 20 km/h	Control calculation and simulation	Met on paper: 6.3 to 6.6 N; loop stable for G = 2 to 6 with the 0.7 Hz filter
R5	Stay within pedal-assist limits	250 W rated continuous motor output; assist only while the drawbar is in tension; no assist above 25 km/h; no throttle	Datasheets; controller settings	Met by design
R6	Trailer brakes itself	With the combination slowing at 3 m/s ² , the trailer pushes the bicycle with 100 N or less; assist cut within 100 ms of drawbar compression above 20 N	Braking calculation; later bench test	At risk: 93 N dry and about 111 N with wet pads on 180 mm rotors with metallic pads (CGM-DDR-002); rotor heating on long descents; cut about 60 ms on paper
R7	Hitch to common bicycles	Fits rear wheels with a 9 mm quick release or a 12 mm thru-axle, 130 to 148 mm spacing, 26 in to 700c; hitch and unhitch in 30 s or less; no wiring to the bike	Hitch design review; later fit survey	At risk: axle stresses pass; thru-axles need a hitch axle per thread pitch and length; nutted axles, some hub gears and some rear-motor e-bikes need adapters
R8	Light enough to handle	Empty trailer 47 kg (104 lb) or less, including the pack: a hard cap for the first prototype only, not a product target (relaxed from 35 kg to 40 kg by CGM-DDR-001 D3, to 45 kg by CGM-DDR-002, then to 47 kg by CGM-DDR-003 A1 on 2026-10-02)	Mass estimate; weighing at TRL 4	Met on paper: 46.3 kg (43.5 kg without side boards); if the weighed trailer is over 47 kg, straps or mesh replace the side boards
R9	Fit bike paths and doors	Overall width 1,000 mm or less; length 2.6 m or less hitched from the bike's rear axle	Model check	Met on paper: 960 mm wide (972 mm over the motor cable), 2.52 m long
R10				

ID	Requirement	Target	Verification (TRL 3 or later)	Status (CGM-CAL-001)
	Stable hitch load	Downward load on the hitch 3 to 10 kg with the design payload centered	Mass and balance estimate	Met on paper: 7.5 kg; holds only for a payload center 32 mm ahead to 57 mm behind the deck center
R11	Weather and temperature	Electronics IP65; motor IP54 or better; operate -10 to 40 °C; the BMS blocks charging below 0 °C	Datasheets and design review	Not verifiable at TRL 3
R12	Affordable	Prototype parts within the \$1,000 value-engineering target, including charger, equal to <code>budget_usd</code> (a hypothetical control target, not a limit; redefined, CGM-DDR-001 D1)	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	Over the value-engineering target by \$4: \$1,004 estimated
R13	Fail safe	Loss of the load cell signal, a reading out of range, a stuck value or a controller fault sets motor current to zero within 100 ms; the trailer stays towable as an unpowered, braked trailer	Fault tree and design review	Not verifiable at TRL 3 (60 ms fast path on paper)
R14	Be seen	Rear red lights and reflectors, side reflectors, and a flag at 1.5 m or more above the ground	Design review	Met by design: flag top 1,570 mm; amber drawbar reflector added (2026-10-02)

■ Assumptions

- Rolling resistance coefficient 0.010 for 20 x 2.15 in tyres on pavement at design load; extra drag area of the loaded trailer 0.1 m², since it rides partly in the rider's wake.
- Air density 1.2 kg/m³; gravitational acceleration 9.81 m/s².
- Drive efficiency from a motor model at each operating point (CGM-CAL-001, section C) rather than a fixed 75 %; charger efficiency 90 %; cell charging efficiency 96 % for LiFePO₄.
- Usable pack energy 90 % of the nominal 384 Wh.
- 250 W and 25 km/h follow the EU pedal-assist exclusion (Regulation (EU) No 168/2013); the US federal definition allows more (under 750 W, 20 mph), so the EU limits are the tighter case. Whether a motorized trailer falls under either rule is unsettled in both the EU and the US. Amish decided on 2026-10-02 (CGM-DDR-001, O2) that the trailer is for private ground only until a written legal check exists for the first partner's country, and that the EU limits stay the design rule as the strictest common case.
- Brake and hitch loads follow the voluntary cycle trailer standards EN 15918 and ASTM F1975, which limit unbraked trailers to 60 kg and 45.4 kg; CargoMule exceeds both, so its brakes are required, not optional.